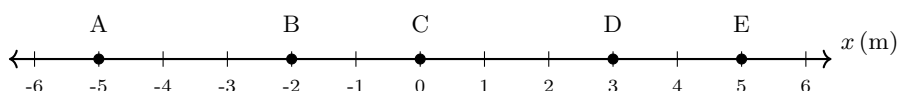


Instructions: Answer all questions. Write in complete sentences unless told otherwise, and show your work for calculations.

Position, Distance, and Displacement

- **Position x :** the location of an object on a coordinate system, measured from the **origin**.
- On a horizontal axis, positive x is right/east and negative x is left/west.
- On a vertical axis, positive y is up/north and negative y is down/south.
- **Distance d :** the total path length (scalar; never negative).
- **Displacement $\Delta x = x - x_0$:** the change in position (vector).
- Same-direction vectors add; opposite-direction vectors subtract. The resultant is a single vector with its own magnitude and direction.

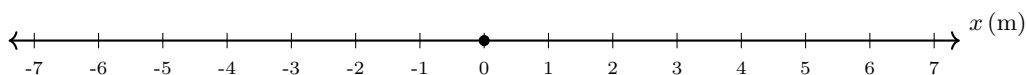
1. The number line below shows the positions of five points, A–E. Write the position of each point, including its sign and unit.



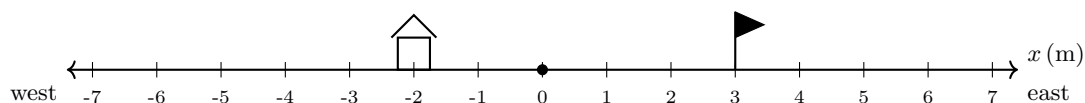
- (a) A: _____
 (b) B: _____
 (c) C: _____
 (d) D: _____
 (e) E: _____

2. Plot and label these positions on the number line:

$$x_1 = -5 \text{ m}, x_2 = -2 \text{ m}, x_3 = 0 \text{ m}, x_4 = +3 \text{ m}, \text{ and } x_5 = +6 \text{ m}.$$



3. A flag is planted at 3 m east and a house sits at 2 m west. Express the given position relative to the indicated landmark.

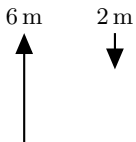


- (a) Position at the house = _____ of the flag.
 (b) Position at the flag = _____ of the house.
 (c) 6 m east of the house = _____ of the flag.
 (d) 1 m east of the house = _____ of the flag.
 (e) 9 m west of the flag = _____ of the house.
 (f) 4 m east of the flag = _____ of the house.

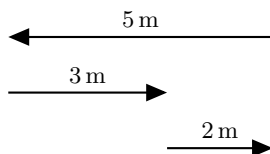
4. Add the two collinear vectors shown. Draw the resultant vector and label the magnitude and direction.



5. Add the two collinear vectors shown. Draw the resultant vector and label the magnitude.



6. Add the three collinear vectors shown. Are you able to draw the resultant vector? Why or why not?

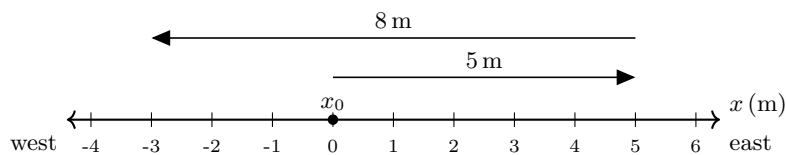


7. Use the equation $\Delta x = x - x_0$ to find each displacement. Include the direction (or the sign) in your answer.

(a) A ball rolls from $x_0 = +2\text{ m}$ to $x = +9\text{ m}$.

(b) A cart moves from $x_0 = +5\text{ m}$ to $x = -3\text{ m}$.

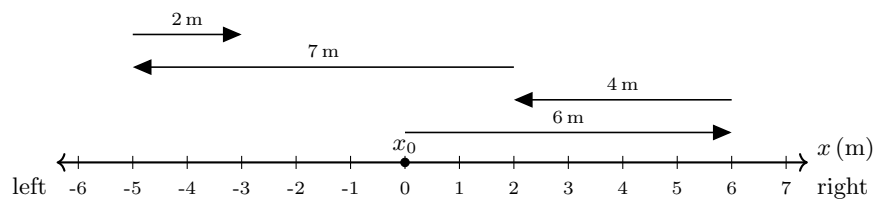
8. A student walks 5 m east, then turns around and walks 8 m west, as shown.



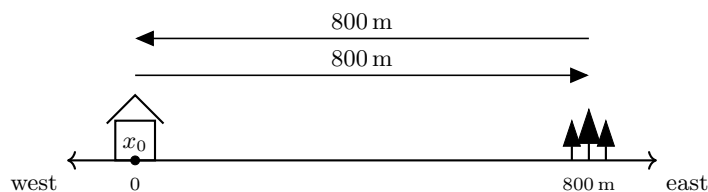
(a) What is the total distance traveled?

(b) What is the total displacement? Include the direction.

9. A remote-controlled car moves along a straight track. Starting from the 0 m mark, it moves +6 m, then -4 m, then -7 m, then +2 m.

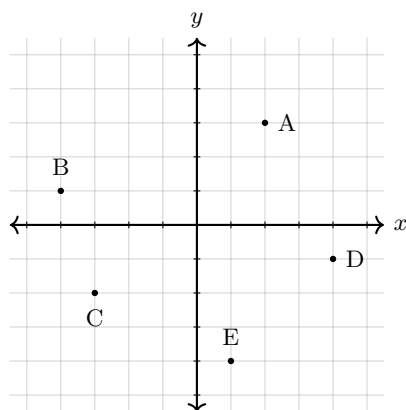


- (a) What is the total distance traveled?
- (b) What is the resultant displacement? Include the direction.
- (c) What is the car's final position?
10. A runner jogs from her house to a park 800 m east, then jogs back home along the same path.



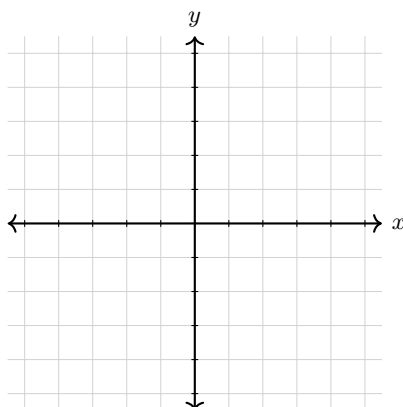
- (a) What is the total distance she travels?
- (b) What is her displacement for the whole trip? Explain.
11. Answer the following about the SI units of distance and displacement.
- (a) What is the SI unit of distance? _____
- (b) What is the SI unit of displacement? _____
- (c) Both quantities use the same unit. Why must a displacement also include a direction?

12. The grid below shows an x - y coordinate plane. Write the ordered pair (x, y) for each labeled point.

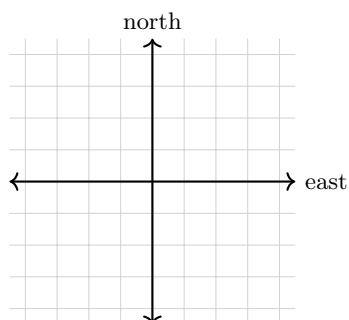


- (a) A: _____
 (b) B: _____
 (c) C: _____
 (d) D: _____
 (e) E: _____

13. Plot and label these points on the coordinate grid: $A(3, 2)$, $B(-4, 1)$, $C(-2, -3)$, $D(5, -2)$, and $E(0, 4)$.

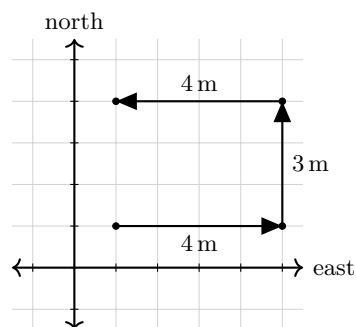


14. On the grid below, east is $+x$ and north is $+y$. Complete each conversion between words and ordered pairs. (One square = one meter.)



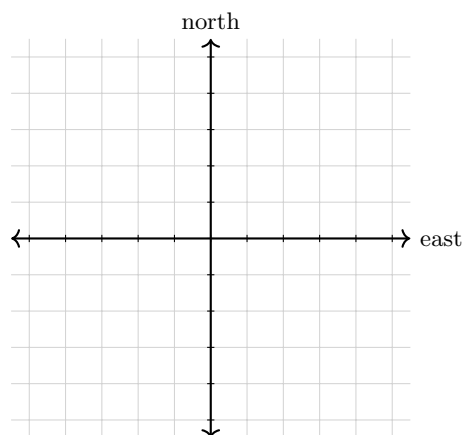
- (a) 3 m north and 2 m east = _____
 (b) 5 m north and 4 m west = _____
 (c) 2 m west and 6 m south = _____
 (d) $(2, -3)$ means _____
 (e) $(-5, -1)$ means _____
 (f) $(-1, 4)$ means _____

15. A robot starts at 1 m east and 1 m north and follows the path shown: 4 m east, then 3 m north, then 4 m west. (One square = one meter.)



- (a) What is the total distance traveled?
 (b) What is the total displacement? Give the magnitude and direction.

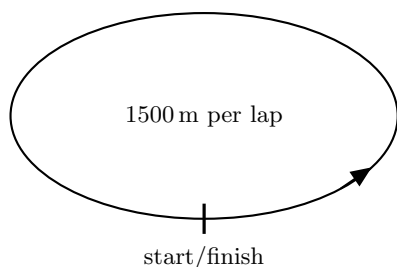
16. An ant starts at position 4 m west and 2 m north and crawls 4 m south, then 5 m east, then 4 m north. Use the blank grid to help you track the ant's position. (One square = one meter.)



(a) What is the total distance traveled?

(b) What is the total displacement? Give the magnitude and direction.

17. A car travels 3 complete laps around a 1500 m oval track and returns to the start/finish line.

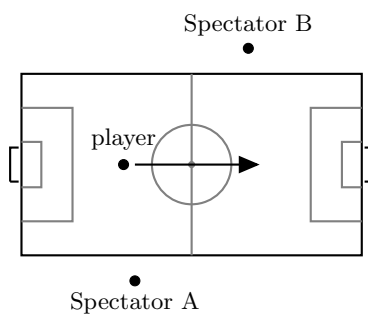


(a) What is the total distance the car travels?

(b) What is the car's total displacement? Explain.

Exit Ticket

18. Two spectators sit on opposite sides of a soccer field, facing each other. A player runs across the field. Spectator A says the player moves to the right; Spectator B says the player moves to the left.



Explain how both spectators can be correct, and give one reference point they could use.